Gemma-2-2B target-side English→French overshoot

SSAE models

SAE models

Cosine to original French activation ↑

French-classifier retention ↑

Within-French geometry correlation ↑

Off-language-axis relative drift ↓

Multiplier m in French + m(αd)

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Purple/teal encode SSAE/SAE; random ribbons show actual draw dispersion. The broad baseline stroke is a reference, not uncertainty.

Semantic panels are activation-space proxies; mean-difference off-axis drift is zero by definition.

- SSAE Linear (constrained L1)
- SSAE Linear (constrained Step-L0)
- SSAE Linear (constrained sigmoid-L0)
- SSAE ReLU (penalized)
- SSAE JumpReLU (penalized)
- SSAE TopK (penalized)
- SAE ReLU (penalized)
- SAE JumpReLU (penalized)
- SAE TopK (penalized)
- Mean-difference control
- Matched-norm random mean ± population SD
- Unsteered baseline reference